Response to the Reviewers (2nd review)

Paper ID 20266918: Decoding Pre-Response EEG Signatures of Choice, Accuracy, and Confidence in Perceptual Decision-Making

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We thank the reviewers and the editor for careful and constructive comments. Every comment led to a re-analysis or a revision. All modifications and additions in the revised manuscript are marked in red font; a clean copy is also provided. Section, table, and figure numbers below refer to the revised manuscript. All numbers in the revised manuscript are generated from one machine-readable result file by the scripts of the reproducibility package.

Summary of the main changes

- All analyses were re-run with one exact solver and one authoritative result file. The inconsistencies the first reviewer found (Tables 4 and 5, Section 5.2, the two p-values of the 1–3 versus 7–9 contrast) are traced to their causes and removed.
- New analyses: causal and truncated-data sensitivity of the preprocessing, flexible (spline) RT control, EEG-plus-RT versus RT-only incremental tests, RT-matched subsampling, training-only and calibration-run normalization, comparison decoders with documented settings and equivalence tests, component baselines, stimulus-locked evaluation, scalp-pattern analysis, per-participant class counts and uncertainty for every calibration contrast, bias-reduced GEE with a participant bootstrap that re-fits the decoder, and minimum-detectable-effect analyses.
- The claims were narrowed. The paper now describes weak, retrospective information in slower trials. Accuracy and confidence information did not survive flexible RT control, and the choice signal did not survive matching on RT and stimulus direction. The protocol is no longer called leakage-controlled.
- The literature was extended (2024 CPP and confidence studies, online EEG confidence control, cross-participant confidence prediction), with a component baseline on the same trials and splits.
- Conflicts of Interest, Author Contributions, Data and Code Availability, and Acknowledgments were added. The layout follows the IJIES format.

Two limits of this revision are stated here as well as in the manuscript. (1) The causal and truncated-data re-extraction from the 1 kHz recordings was completed for 9 of the 24 participants, because the 48 recordings of about 800 MB each could not all be re-processed within the revision period; the decimation step is shown to be inert in this subset, whereas the effect of the dataset's own band-pass could not be isolated (Section 5.4). (2) The proposed public repository has not been created; the package is ready for release, and the repository address in the Data and Code Availability statement is left for the authors.

Reviewer 1

Comment 1.1: temporal support of preprocessing

> *Establish the temporal support of preprocessing in Sections 3 and 4.1 before describing the protocol as leakage-controlled. Specify the phase, order, impulse response, and boundary handling of the supplied band-pass filtering and the additional low-pass decimation, including whether any operation uses samples after the feature-window endpoint. A 150 ms buffer alone does not demonstrate isolation from response-related activity. Provide a causal-filter or appropriately truncated-data sensitivity analysis and quantify any change in the primary AUCs; do not assume that contamination occurred without testing it.*

Response. We agree that a buffer alone does not demonstrate isolation. We specified every operation that precedes the window average and its temporal support, and we tested the part that can be tested. The

decimation filter we applied has zero phase, and its analytic impulse response places 0.25% of its absolute weight after the window end and almost none beyond the buffer. In the 9 participants that could be re-extracted from the 1 kHz data, a plain 1 kHz mean, a causal decimation filter, and decimation of data truncated at the window end changed the AUCs by at most 0.007 on average. The band-pass filter supplied with the dataset is not documented, and we cannot re-run it. A zero-phase band-pass of the usual form would put a substantial part of its weight after the window end, so we do not claim that leakage is excluded. Our proxy band-pass variants change the features drastically (correlation with the primary features 0.16 to 0.26 for the causal and truncated forms), and they therefore do not isolate leakage; we say so explicitly. The split of a zero-phase filter into its inside-window and after-window parts is implemented, and it is listed as future work because the re-extraction could not be completed.

Changes in the manuscript. Section 4.1 (the table of operations with phase and support is in the Supplementary Material), Section 5.4 with Table 4 and the Supplementary Material, Supplementary Table S9, Section 7. The term leakage-controlled was removed from the abstract, the introduction, and the conclusion.

Comment 1.2: reconciling Tables 4 and 5 and Section 5.2

> *Reconcile the primary mean-amplitude results across Tables 4 and 5 and Section 5.2. Logistic choice is 0.526 in Table 4 but 0.527 in Table 5, and confidence is 0.540 versus 0.541. Shrinkage-LDA choice is 0.526 in Table 4 and Section 5.2 but 0.525 in Table 5. The spatiotemporal accuracy result is 0.539 in Section 5.2 but 0.541 for logistic regression with four bins in Table 5. Identify any different trial sets, implementations, or aggregation rules, or regenerate these entries from one authoritative result file. The stated common trials, window, and splits do not explain the discrepancies.*

Response. The differences were caused by two things: the standardization population and the solver. In the first manuscript the feature standardization of one table used all valid trials of a participant (including the excluded fast trials), and the other used the retained trials; in addition, scikit-learn solvers stopped at different tolerances. We regenerated every entry from one result file with one exact Newton solver, validated against scikit-learn to a mean AUC difference below 0.0002. The two normalization populations are now both reported: choice 0.527 (retained trials) versus 0.526 (all trials), accuracy 0.535 versus 0.535, confidence 0.540 versus 0.540. The numbers in the text, the tables, and the figures now come from the same file, and no table entry is typed by hand.

Changes in the manuscript. Table 2, Section 5.2, Section 5.5 with the Supplementary Material, and Supplementary Tables S5 and S10. The scripts and the result file are in the reproducibility package.

Comment 1.3: p-values of the 1–3 versus 7–9 contrast and of Table 7

> *The ratings 1-3 versus 7-9 contrast among incorrect trials has the same pooled AUC of 0.487 but different p-values: 0.581 in Section 5.6 and Table 6, and 0.621 in Table 7. Reconcile the permutation runs and seeds or explain a distinct statistic or null hypothesis. Also audit the Table 7 p-values against the stated 500-permutation formula in Section 4.4. With denominator 501, values rounding to 0.003, 0.053, 0.336, and 0.023 are unattainable at three decimal places. If these tests used another permutation count, a different test, or adjusted p-values, specify that procedure rather than silently applying the primary inference description.*

Response. The two p-values for the incorrect-trial contrast came from two permutation runs with different numbers of permutations and different seeds (500 and 300), and the first manuscript described only the 500-permutation procedure. The reviewer is right that the p-values of the other table cannot be produced with the denominator 501. They were produced with 300 permutations (denominator 301), so that 0.003, 0.053, 0.336, and 0.023 are attainable. All permutation tests were re-run with 1,000 permutations, with seeds, the permutation statistic, and the null mean and 95th percentile stored for every test. The incorrect-trial contrast is

reported once, and the formula p = (1 + k) / (1 + n) with n = 1,000 is stated in Section 4.3. Inference that is not based on permutations (stimulus-locked and time-resolved analyses) is labelled as one-sided t-tests with Holm adjustment, and no value is described as a permutation p-value unless it is one.

Changes in the manuscript. Section 4.3, the Supplementary Material, Section 5.8 with Table 5 and the Supplementary Material, Section 5.6, and Supplementary Tables S6 and S12.

Comment 1.4: timing descriptions

> *Correct the timing descriptions against the implemented windows. A 600 ms window ending 150 ms before the response spans -750 to -150 ms, not the 600-to-150 ms interval stated in the Conclusion. In Section 5.5, the three significant accuracy-window centers are listed as -600, -550, and -400 ms, although the text places them between 0.65 and 0.45 s before the response. The -400 ms center is outside that interval. State whether boundaries or centers are intended and report the exact significant intervals consistently with Fig. 5.*

Response. The implemented window is 600 ms long and ends 152 to 155 ms before the response (the response time is rounded down to the 4 ms sampling grid), so it spans about −755 to −152 ms; the conclusion now says so. In the time-resolved analysis the windows are 100 ms long. The reported significant windows for accuracy start at −650, −600, and −450 ms, that is, they span −650 to −550, −600 to −500, and −450 to −350 ms before the response; centers and boundaries are no longer mixed, and the intervals are given in the text, in Fig. 2, and in Supplementary Table S12. These three windows are significant after Holm adjustment, none remains after flexible RT residualization, and the time-resolved analysis is stated not to validate the buffer's physiological rationale.

Changes in the manuscript. Section 4.1, Section 5.6, Section 8, the abstract, and Fig. 1.

Comment 1.5: RT-control interpretation

> *Revise the RT-control interpretation in Section 6.1. Accuracy does not vanish after RT stratification: Section 5.2 reports AUC 0.517 and p = 0.030 within RT terciles, whereas linear residualization gives 0.503 and p = 0.415. These controls answer different questions, and coarse terciles retain within-bin RT variation. Report corrected and uncorrected inference explicitly and assess incremental prediction using RT-only versus EEG-plus-RT models under the same held-out-participant protocol, with a flexible RT control or matched-trial sensitivity analysis.*

Response. We agree. Accuracy information does not vanish under every control. We now report, side by side and with the correction made explicit, the uncorrected result, linear and 5-degree-of-freedom spline residualization, RT-tercile stratification, an RT-only model, an EEG-plus-RT model against the RT-only model with a label permutation that is stratified by RT decile, and matched-trial subsampling. Within RT terciles the accuracy AUC was 0.515 (permutation p = 0.106); the earlier p = 0.030 was a one-sided t-test on scores normalized differently, and the present analysis does not reproduce a significant effect after correction. EEG added nothing to RT for accuracy (difference -0.010) or confidence (-0.021), but it added +0.019 to RT for choice (permutation p = 0.010). The conclusions are written as dependent on the adjustment, and the text states that terciles keep within-bin RT variation.

Changes in the manuscript. the Supplementary Material, Section 5.3 with Table 3, Section 6.2, the abstract, and Section 8.

Comment 1.6: window grid and tie-breaking rule

> *Specify the complete window grid and the tie-breaking rule in Section 4.1. Maximizing L subject to B >= 100 ms and at least 80% retention does not uniquely choose B = 150 ms: for the same L = 600 ms, B = 100 ms must retain at least as many trials. State the additional criterion used to choose the buffer, show*

its implementation and retention counts, and distinguish the original plan from later amendments. Do not infer preregistration from an undated internal plan.

Response. The reviewer is correct. B = 100 ms retains more trials than B = 150 ms for the same L, so the stated rule does not select B = 150 ms. The window was chosen in a second step of the first plan version: the largest L with at least 80% retention at B ≥ 100 ms and, among the windows that tie, the largest B. We now state this as the rule, give the full grid (L from 200 to 800 ms, B from 0 to 350 ms) with retention percentages and trial counts, and show that no window of the grid was selected from the decoding results. We state that the plan is an undated internal document that was edited in place, that it was not preregistered, and which parts were added later (comparison decoders as an amendment, RT residualization and time-resolved analyses afterwards). The file-system times are listed as the only available evidence of order.

Changes in the manuscript. Section 4.1, the Supplementary Material with the Supplementary Material, Section 4.5, and Supplementary Tables S1 and S2.

Comment 1.7: normalization and deployment setting

Define the normalization and deployment setting precisely in Sections 4.2 and 4.5. Both feature standardization and RT regression use the held-out participant's unlabeled trial distribution. This is transductive adaptation rather than label leakage, but it requires a batch of target-participant data and, for the RT control, completed response times. State which trials enter each fit, how medium-confidence trials and excluded trials are handled, and add a training-only or separate-calibration sensitivity analysis if prospective prediction in a new participant is claimed.

Response. We agree that feature standardization and RT regression based on the target participant's unlabeled batch is transductive adaptation. The revised text defines which trials enter each fit (retained trials of the participant, labels never used, medium-confidence trials excluded from the confidence target only and not from standardization) and states that the setting needs a batch of target data and, for the RT control, completed response times. We added sensitivity analyses with training-only normalization (choice 0.524, accuracy 0.537, confidence 0.544), a separate calibration run (0.525, 0.534, 0.536), and causal running normalization (0.513, 0.544, 0.540). The choice effect weakens with causal running normalization, which the text reports. Claims about prospective prediction in a new participant are limited accordingly.

Changes in the manuscript. the Supplementary Material, Section 5.5 with the Supplementary Material, Section 6.3.

Comment 1.8: comparison decoders and the log-variance AUCs

Document the target coding and implementation of the comparison decoders in Section 4.3, including positive-class score orientation, solver, convergence, RBF kernel scale, MLP validation selection, and random seeds. Section 5.3 dismisses below-chance log-variance AUCs as a known feature-scale artifact without demonstrating that mechanism, despite within-participant standardization. Inspect label orientation and participant distribution shifts, report relevant diagnostics and permutation results, and distinguish failure to outperform the primary model from equivalence or absence of information.

Response. The target coding (positive class: rightward choice, correct, high confidence), the score orientation (decision function of the positive class), solvers, convergence tolerances, the RBF scale (gamma = 'scale'), the MLP validation split, and all seeds are now listed in the text and in a supplementary table. The explanation of below-chance log-variance AUCs as a scale artifact was removed. We inspected label orientation and distribution shift and report diagnostics: the mean log-variance AUCs are 0.476 (accuracy) and 0.471 (confidence), with lower-tail permutation p-values of 0.024 and 0.048, a negative correlation between the held-out and the training pattern in most participants, and chance-level within-participant cross-validation.

These values are consistent with a training-set pattern that does not transfer in sign, and we describe them as below-chance transfer of unclear origin and not as a known artifact. Differences from the primary decoder are tested with an equivalence margin of ±0.02 AUC; classifiers are called equivalent only when the 90% interval lies inside the margin, and otherwise inconclusive, and not as equal or as showing absence of information.

Changes in the manuscript. the Supplementary Material, the Supplementary Material with the Supplementary Material and the Supplementary Material, and Supplementary Table S4.

Comment 1.9: checkable calibration and sequential inference

> *Make the calibration and sequential inference independently checkable. For every calibration contrast, report class counts per held-out participant, the number of evaluable participants, missing-class handling, the aggregation statistic used in permutation tests, and uncertainty intervals. In Section 4.10, write the reported-confidence benchmark and the RT/accuracy outcome models explicitly, specify their links and covariates, and report the handling of generated decoder scores. With only 24 participant clusters, use an appropriate small-sample correction or participant-level bootstrap that refits the decoding pipeline to check the GEE conclusions.*

Response. For each calibration contrast we now report the class counts, the number of participants with trials, the number of evaluable participants (both classes present with the rule stated), the missing-class handling, the aggregation statistic (mean participant AUC as primary, pooled AUC as secondary), the permutation scheme (labels shuffled within participant inside the contrast subset, whole procedure re-fitted, 1,000 permutations), and bootstrap intervals over participants; per-participant counts are in a supplementary table. For example, the incorrect-high versus incorrect-low contrast has 853 trials from 24 participants, of which 17 are evaluable, with mean participant AUC 0.500 [0.438, 0.557]. In the sequential analysis the benchmark, the RT model, and the accuracy models are written as equations with their links and covariates, the decoder scores enter as out-of-fold values with their generating process stated, and inference uses bias-reduced (Mancl and DeRouen) covariance with a t(23) reference and a participant bootstrap that re-fits the decoder. The conclusions of the GEE agree with those of the bootstrap, except for the reported-confidence benchmark whose coefficient is positive in both.

Changes in the manuscript. the Supplementary Material, the Supplementary Material, Section 5.8 with Table 5, Section 5.9 with Table 6, and Supplementary Tables S6 and S7.

Comment 1.10: reproducibility package

> *The manuscript provides a code link, but the reported inconsistencies and analysis-plan amendments require a versioned reproducibility package tied to the submitted results. Include the original plan and amendments with dates, the exact release/commit and environment, channel mask, retained-trial and adjacent-pair indices, normalization and RT-regression settings, decoder configurations, seeds and selection rules, participant-level predictions, and machine-readable permutation statistics and table-generation outputs. These artifacts are needed to reproduce the window selection, distinguish alternative runs, and verify all reported AUCs and p-values.*

Response. A versioned package was assembled and tagged locally. It contains the analysis plan and amendments with their file-system times, the environment (library versions and hardware), the channel mask, the retained-trial and adjacent-pair indices, the normalization and RT-regression settings, all decoder configurations, seeds and selection rules, participant-level predictions, permutation statistics, and the scripts that generate every table and figure from one result file with hashes. The repository address is left for the authors to complete, since the release itself is theirs to make.

Changes in the manuscript. Section 4.5 and the Data and Code Availability statement; Supplementary Table S1.

Reviewer 2

Comment 2.1: literature framing and component baseline

> *The conceptual contribution can be useful as a careful secondary analysis of weak, transferable EEG information, but the literature framing is incomplete. Relevant work published in 2024 has examined predecision CPP-confidence relationships while controlling accuracy, RT, and evidence quality, and EEG-based online confidence control in dynamic perceptual environments. Earlier work also studied subject- and task-independent confidence prediction. Discuss how the present response-buffered, out-of-participant setting differs in features, normalization, task, and target. Where feasible, add a justified component/slope baseline evaluated on the same retained trials and splits; do not treat cross-study AUCs from different protocols as a fair ranking.*

Response. We added a subsection that compares the present study with the 2024 work on pre-decision CPP and confidence, the 2024 EEG-based online confidence control, and earlier subject- and task-independent confidence prediction, along the dimensions features, normalization, task, and target, and the comparison table of the closest studies (moved to the Supplementary Material). We state that AUCs of different protocols are not ranked. A CPP-like (mean and slope) and a lateralized-readiness-like baseline were evaluated on the same retained trials and splits as the primary decoder. For choice neither component set exceeded chance, whereas for confidence the two CPP-like features came close to the whole-scalp decoder, an effect that is itself RT-dependent.

Changes in the manuscript. the Supplementary Material with the Supplementary Material, the Supplementary Material, the Supplementary Material with the Supplementary Material.

Comment 2.2: retrospective decoding versus online prediction

> *A retrospective window defined by each trial's eventual response time can identify pre-response associations, but it is not automatically a deployable advance prediction system: its endpoint cannot be located prospectively without knowing when the response will occur. Together with whole-participant unlabeled normalization, this limits the meaning of transfer to a new participant. Separate retrospective decoding from online prediction throughout the claims. If practical advance prediction is intended, demonstrate a stimulus-locked or causally updated evaluation and account for processing latency and calibration requirements.*

Response. We agree and have separated the two throughout. The primary result is described as retrospective decoding in a window defined by the response time, which cannot be placed in advance. We added a stimulus-locked evaluation with a 300 ms window at several latencies and with a 0 or 150 ms processing delay, and a causal running normalization. Of 48 combinations of decision time, latency, and target, only one survived Holm correction, so the retrospective effects do not reappear in advance, and the claims about advance prediction were removed. Calibration data requirements and processing latency are discussed.

Changes in the manuscript. Section 4.4, Section 5.6 with Fig. 2 and the Supplementary Material, Supplementary Table S8, Section 6.3, the abstract, and Section 8.

Comment 2.3: RT control is not a causal test

> *The response-time controls are a strength, yet removing an RT association from EEG is not a causal demonstration that the signal is merely speed contamination. RT can share the decision-generating process with neural features, and residualizing those features can also remove meaningful decision*

information. The remaining accuracy effect under tercile stratification further prevents a uniform conclusion that both effects are fully explained by RT. Frame the evidence as dependence on the chosen adjustment and test whether EEG adds information beyond flexible behavioral predictors before drawing a stronger mechanistic conclusion.

Response. We agree. The discussion now frames the results as dependent on the adjustment, notes that RT can share the decision process with the neural features, and that residualization can remove decision information. To test whether EEG adds information beyond flexible behavioral predictors we used the EEG-plus-RT versus RT-only comparison with spline RT terms, and a decoder trained on RT-residualized features. The tercile result is reported next to the others.

Changes in the manuscript. the Supplementary Material, Section 5.3, Section 6.2.

Comment 2.4: power, equivalence, and nulls

The rare-cell feasibility rule establishes that an analysis is possible, not that it has enough power to exclude meaningful calibration effects. Likewise, a nonsignificant difference between classifiers does not establish equal performance. Add participant-level uncertainty intervals and a justified smallest effect of interest or power/sensitivity analysis for the main null claims, including calibration, classifier comparisons, and sequential effects. The permutation null's 95th percentile is a rejection threshold, not by itself an estimate of detection power. Preserve the distinction between no detected effect and evidence of absence.

Response. The rare-cell rule is now described as a feasibility rule only. For every null claim we report participant-level intervals and the minimum detectable effect at 80% power (Hanley and McNeil approximation for pooled contrasts, permutation null plus 0.84 standard errors for mean AUCs, and bias-reduced coefficients for the GEE), and we use a smallest effect of interest of AUC 0.55 and an equivalence margin of ±0.02 for classifier comparisons. We state that the 95th percentile of the null is a rejection threshold and not a power estimate. Nulls are written as no detected effect, and equivalence is claimed only where the interval supports it.

Changes in the manuscript. the Supplementary Material, Section 5.10, the Supplementary Material, Section 6.4, and Supplementary Table S11.

Comment 2.5: interpretation of the choice signal

The interpretation of the surviving choice signal remains underdetermined because stimulus direction, choice, and response preparation are coupled. The within-direction AUCs and mixed significance provide a useful partial control, but neither establish a choice-specific neural representation nor justify exclusion of motor preparation by a fixed buffer. A joint control with RT matching, clearly corrected inference, and component or scalp-pattern analyses would strengthen interpretation. Otherwise, keep the conclusion at the level of weak discriminative information in this task and acknowledge that the time-resolved analysis did not validate the buffer's physiological rationale.

Response. We added a joint control that matches trials within participant on RT decile and stimulus direction; the choice AUC was 0.506 in that analysis, and we state that the retained information is therefore not shown to be independent of RT and stimulus direction. We added within-direction decoding, a scalp pattern analysis (Haufe activation patterns) with comparison against a left/right-lateralized template, and a time-resolved analysis whose result is stated as not validating the buffer's physiological rationale. The conclusion is kept at the level of weak discriminative information in this task, without claiming a choice-specific representation or an exclusion of motor preparation.

Changes in the manuscript. Section 5.7 with the Supplementary Material, Section 5.3, Section 6.1, and Section 6.1.

Comment 2.6: scope and per-participant retention

> *The manuscript already acknowledges selection of slower trials and the weakness of the confidence proxy, which should remain central to its scientific scope. Quantify per-participant retention and uncertainty, and avoid extending these results to fast decisions, ordinary out-of-scanner EEG, or individual overconfidence detection. In particular, the sequential null concerns this weak, largely RT-related decoder score and does not test the absence of a neural confidence influence on later behavior. A stronger confidence readout or independent replication would be required for that broader inference.*

Response. We now give per-participant retention counts and percentages (median 97%, minimum 24%, with 4 participants retaining less than half of their trials) and compare retained and excluded trials. The scope statements say that the results concern slower decisions, scanner EEG, and a weak confidence proxy, and that they do not extend to fast decisions, ordinary EEG outside the scanner, or individual overconfidence detection. The sequential null is described as concerning the weak, largely RT-related decoder score, and not as evidence about the absence of a neural confidence influence on later behavior.

Changes in the manuscript. Section 5.1 with Table 1, Section 5.9, Section 6.4, Section 6.4, and Supplementary Table S3.

Editor

- Red font: all modifications and additions are in red in Revised_Manuscript_R1_marked; Revised_Manuscript_R1_clean is the same text without marking.
- Conflicts of Interest and Author Contributions were added as sections; the content of both is to be confirmed by the authors, because it concerns personal facts we cannot state on their behalf.
- Code release: a public repository with the training and evaluation scripts, preprocessing, configurations, seeds, and the result file is prepared as a package; the repository address is to be inserted by the authors.
- Format: text 11 pt and tables 10 pt; figures of one-column width, none side by side, none with (a), (b), (c) labels or an outer border, and all figure text at least 10 pt; equations 11 pt with italic variables, numbered at the right margin, with no asterisk as a multiplication sign; each table and figure is at the top or bottom of a column and is not divided across pages; references follow the form 'Vol.35, No.8, pp.3478-3484, 2020' with city and country for conference proceedings and with bracketed numbers, ranges in one bracket; author names and the affiliation are written in full without abbreviations or acronyms, the affiliation is not repeated, and titles are removed.
- Length: the revised manuscript was shortened to 11 pages. Descriptive tables and figures (operations, decoder comparison, component baselines, window grid, confidence definitions, rating and AUC distributions, scalp patterns, stimulus-locked and two-stage figures) were moved to the Supplementary Material, and the text was condensed without removing any response to a reviewer comment.